

Problem

Let

$$R \in \text{SO}(3), \quad p \in \mathbb{R}^3.$$

Calculate

$$\boxed{\frac{\partial(R^{-1}p)}{\partial R}}$$

using both right and left perturbation models.

1 Right perturbation

Use

$$R' = R \text{Exp}(\delta\theta).$$

1.1 Method 1: Chain rule

Consider

$$R \mapsto R^{-1} \mapsto R^{-1}p.$$

Then

$${}^R J_R^{R^{-1}p} = {}^{R^{-1}} J_{R^{-1}}^{R^{-1}p} {}^R J_R^{R^{-1}}.$$

For $S \in \text{SO}(3)$, the right-perturbation Jacobian of Sp is

$${}^S J_S^{Sp} = -Sp^\wedge.$$

Setting $S = R^{-1}$,

$${}^{R^{-1}} J_{R^{-1}}^{R^{-1}p} = -R^{-1}p^\wedge.$$

For inversion under a right perturbation,

$${}^R J_R^{R^{-1}} = -\mathbf{Ad}_R = -R.$$

Therefore

$${}^R J_R^{R^{-1}p} = (-R^{-1}p^\wedge)(-R) = R^{-1}p^\wedge R.$$

Using

$$(Ra)^\wedge = Ra^\wedge R^T,$$

with R^{-1} in place of R ,

$$R^{-1}p^\wedge R = (R^{-1}p)^\wedge.$$

Hence

$$\boxed{{}^R J_R^{R^{-1}p} = (R^{-1}p)^\wedge \quad (\text{right perturbation}).}$$

1.2 Method 2: Directly from the definition

Choose an arbitrary $u \in \mathbb{R}^3$ and let the perturbation be hu . Then

$${}^R J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{(R \text{Exp}(hu))^{-1} p - R^{-1} p}{h}.$$

Since

$$(R \text{Exp}(hu))^{-1} = \text{Exp}(-hu) R^{-1},$$

we obtain

$${}^R J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{\text{Exp}(-hu) R^{-1} p - R^{-1} p}{h}.$$

Using

$$\text{Exp}(-hu) = I - hu^\wedge + O(h^2),$$

$${}^R J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{-hu^\wedge R^{-1} p + O(h^2)}{h}.$$

Since

$$a^\wedge b = -b^\wedge a,$$

$$-u^\wedge R^{-1} p = (R^{-1} p)^\wedge u.$$

Thus

$${}^R J_R^{R^{-1}p} u = (R^{-1} p)^\wedge u.$$

Because u is arbitrary,

$$\boxed{{}^R J_R^{R^{-1}p} = (R^{-1} p)^\wedge.}$$

2 Left perturbation

Use the left-plus convention

$$R' = \text{Exp}(\delta\theta) R.$$

2.1 Method 1: From equation (57)

The left-perturbation result can also be obtained directly from the right-perturbation Jacobian by using equation (57) of the paper:

$$\boxed{{}^\varepsilon \frac{Df(X)}{DX} = \mathbf{Ad}_{f(X)} {}^X \frac{Df(X)}{DX} \mathbf{Ad}_X^{-1}.} \quad (57)$$

For the present exercise,

$$f(R) = R^{-1} p.$$

The right-perturbation Jacobian has already been found to be

$$\boxed{{}^R J_R^{R^{-1}p} = (R^{-1} p)^\wedge.} \quad (11)$$

Applying (57) gives

$${}^\varepsilon J_R^{R^{-1}p} = \mathbf{Ad}_{R^{-1}p} {}^R J_R^{R^{-1}p} \mathbf{Ad}_R^{-1}. \quad (12)$$

The output

$$R^{-1}p \in \mathbb{R}^3$$

belongs to the additive translation group $(\mathbb{R}^3, +)$. This group is commutative, and its adjoint matrix is the identity:

$$\boxed{\mathbf{Ad}_{R^{-1}p} = I.} \quad (13)$$

For $\text{SO}(3)$,

$$\boxed{\mathbf{Ad}_R = R, \quad \mathbf{Ad}_R^{-1} = R^{-1}.} \quad (14)$$

Substituting (11)–(14) into (12),

$${}^\varepsilon J_R^{R^{-1}p} = I (R^{-1}p)^\wedge R^{-1} \quad (1)$$

$$= (R^{-1}p)^\wedge R^{-1}. \quad (2)$$

Using

$$(R^{-1}p)^\wedge = R^{-1}p^\wedge R,$$

we obtain

$${}^\varepsilon J_R^{R^{-1}p} = R^{-1}p^\wedge R R^{-1} \quad (3)$$

$$= \boxed{R^{-1}p^\wedge}. \quad (4)$$

Thus equation (57) converts the right-perturbation result directly into the left-perturbation result:

$$\boxed{{}^R J_R^{R^{-1}p} = (R^{-1}p)^\wedge \implies {}^\varepsilon J_R^{R^{-1}p} = R^{-1}p^\wedge.}$$

This agrees with the result obtained independently from the direct left-perturbation limit.

2.2 Method 2: Directly from the definition

To make the Jacobian limit rigorous, choose an arbitrary direction

$$u \in \mathbb{R}^3$$

and set

$$\delta\theta = hu.$$

Then

$${}^\varepsilon J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{(\text{Exp}(hu)R)^{-1}p - R^{-1}p}{h}.$$

Using

$$(AB)^{-1} = B^{-1}A^{-1},$$

we have

$$(\text{Exp}(hu)R)^{-1} = R^{-1} \text{Exp}(-hu).$$

Therefore

$${}^\varepsilon J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{R^{-1} \text{Exp}(-hu)p - R^{-1}p}{h}.$$

For small h ,

$$\text{Exp}(-hu) = I - hu^\wedge + O(h^2).$$

Hence

$$\varepsilon J_R^{R^{-1}p} u = \lim_{h \rightarrow 0} \frac{R^{-1}(I - hu^\wedge)p - R^{-1}p + O(h^2)}{h} \quad (5)$$

$$= \lim_{h \rightarrow 0} \frac{-hR^{-1}u^\wedge p + O(h^2)}{h}. \quad (6)$$

Using

$$u^\wedge p = -p^\wedge u,$$

we get

$$-R^{-1}u^\wedge p = R^{-1}p^\wedge u.$$

Therefore

$$\varepsilon J_R^{R^{-1}p} u = R^{-1}p^\wedge u.$$

Since u is arbitrary,

$$\boxed{\varepsilon J_R^{R^{-1}p} = R^{-1}p^\wedge.}$$

3 Final comparison

For the right perturbation

$$R' = R \operatorname{Exp}(\delta\theta_R),$$

$$\boxed{\frac{\partial(R^{-1}p)}{\partial R} = (R^{-1}p)^\wedge.}$$

For the left perturbation

$$R' = \operatorname{Exp}(\delta\theta_L)R,$$

$$\boxed{\frac{\partial(R^{-1}p)}{\partial R} = R^{-1}p^\wedge.}$$

These two Jacobians are related by

$$(R^{-1}p)^\wedge = R^{-1}p^\wedge R,$$

and the perturbation coordinates satisfy

$$\delta\theta_L = R \delta\theta_R.$$

Indeed,

$$R^{-1}p^\wedge \delta\theta_L = R^{-1}p^\wedge R \delta\theta_R = (R^{-1}p)^\wedge \delta\theta_R.$$

Thus the two perturbation conventions describe the same first-order variation using different tangent-coordinate representations.

Useful identity

For $R \in \operatorname{SO}(3)$ and $a \in \mathbb{R}^3$,

$$\boxed{(Ra)^\wedge = Ra^\wedge R^T.}$$